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An energy-efficient single-cycle RV32I microprocessor for edge computing applications☆

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ABSTRACT

With the advancement in the Internet of Things (IoT) and Wireless Sensor Node (WSN) applications computation on edge is preferred over computation on the cloud. However, due to limited resource availability in the edge environment, energy efficiency can be seen as the primary barrier in the growth of edge computing devices. In edge computing applications like wearable devices, forest monitoring systems, and other such applications, power is of utmost importance. The use of a complex, pipelined, general-purpose processor is redundant for such low-end applications and it leads to high power consumption. Hence, edge computing systems require low-power solutions while still maintaining a minimum performance level. In this work, an energy-efficient single-cycle RISC-V instruction set architecture (ISA) based RV32I processor is designed for low-end edge computing applications. The design incorporates base integer implementation of RISC-V ISA. The proposed computing system is designed using Verilog HDL and synthesized on Semi-conductor Laboratory (SCL) 180 nm CMOS process technology node. The synthesis tool reports power consumption of 14.7 mW at 50 MHz frequency with an area of 0.24 mm^2. Though synthesis of this design reports area of 0.24 mm^2, the design is taped out as a part of an available shuttle project where die size is fixed to 5 mm × 5 mm. The PNR (Place and Route) tool reports power consumption of 11.8 mW at an operating frequency of 40 MHz. A basic functional testing of proposed fabricated RV32I chip is carried out in laboratory experimental setup and results are presented. The comparison result shows that the proposed RV32I processor consumes 0.29 mW/MHz power which is less than state-of-the-art low-end processors. Hence, the proposed design is suitable for low-end edge computing applications.

1. Introduction

The rise of the Internet of Things (IoT) and Wireless Sensor Node (WSN) applications have enhanced the everyday life of humans by creating a network of interconnected computing, sensing, and monitoring devices. IoT is estimated to hit 500 billion internet-connected devices by 2030 [1–3]. The evolution and advancement of the IoT and WSN applications have increased the demand for edge computing systems. IoT and WSN applications have enabled the explosive growth of sensing and monitoring devices; e.g., wearable devices, smart home appliances, and forest monitoring system. Many IoT and WSN applications use battery-operated power sources. Moreover, most IoT and WSN devices have limited resources for computing, storage, and communication [4–7].

A general-purpose processor [8,9] supporting multiply, division, floating-point, double floating-point, and atomic instructions will have a lot of redundant hardware and it may not be suitable for such resource-constrained applications [10–12]. The deeply pipelined processors [13,14] provide good performance but they are power-hungry. So, they are also not suitable for low-end, energy-efficient applications [15–17].

For such resource-constrained applications, RISC-V (Reduced Instruction Set Computer-V) ISA [18] is the best candidate due to its low power features [19–21]. The modular approach of RISC-V allows a processor to keep only those instructions necessary for a particular low-end application. RISC-V-based processors provide the facility to remove the redundant hardware which leads to a significant reduction in power consumption, making it suitable for low-end IoT and WSN applications [22–24].

Hence, in this paper, an energy-efficient single-cycle RISC-V-based RV32I processor architecture is proposed for low-end edge computing

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	31 25	24 20	19 15	14 12	11 7	6 0
R-type	funct7	rs2	rs1	funct3	rd	opcode
I-type	immediate[11:0]		rs1	funct3	rd	opcode
S-type	immed[11:5]	rs2	rs1	funct3	immed[4:0]	opcode
SB-type	immed[12,10:5]	rs2	rs1	funct3	immed[4:1,11]	opcode
UJ-type	immediate[20,10:1,11,19:12]				rd	opcode
U-type	immediate[31:12]				rd	opcode

Fig. 1. RISC-V instruction encoding format [18].

applications. The design incorporate base integer implementation of RISC-V ISA. Since the design target low-end embedded applications, no standard or non-standard extension is added to the base integer implementation. Further the proposed RV32I processor is designed and fabricated on 180 nm CMOS process technology node.

The rest of the paper is organized as follows. Section 2 introduces the features of RISC-V ISA. Section 3 discusses the design aspects and architecture of RISC-V based single-cycle RV32I microprocessor core. Section 4 presents the HDL implementation and its FPGA prototype. Section 5 presents the ASIC (application-specific integrated circuit) implementation of the proposed RISC-V32I processor core architecture on 180 nm CMOS process technology node. It also presents the comparison of the proposed processor to the commercially-available low end embedded processors. Further, silicon level validation of fabricated chip in laboratory experimental setup is presented in this section. Section 6 concludes the paper.

2. RISC-V ISA features

The RISC-V ISA was developed in the Computer Science Department at the University of California, Berkeley. It is designed to support the research and education in computer architecture and now it is used as a standard open architecture for many industry implementations. In this section, the features of RISC-V ISA that makes it suitable for low-end edge computing applications are presented.

The first and foremost reason of choosing RISC-V ISA is that it is open source which makes it a free ISA. The open source feature reduces the complexity and Intellectual property (IP) issues found in MIPS [25], SPARC [26], and x86 [27] based research projects. There are other open source ISAs like SPARC V8, OpenRISC etc.RISC-V have another major advantage over all other existing ISAs is that it provides modular approach. The modular approach gives the freedom to designers to implement only those set of instructions (along with base integer implementation) which is desired for a particular class of application. So, modular approach allow the computer architects to cut off unnecessary hardware to makes the design efficient.

The RISC-V ISA consists of base integer ISA and optional extensions to the base ISA. The base ISA must be present in any implementation. The optional extensions of RISC-V instruction set are divided into two parts; standard extensions and non-standard extensions. The standard extensions are usually important and a standard extension should not conflict with any other standard extensions. However, a non-standard extension is for a domain-specific task and it may be highly specialized. A non-standard extension may conflict with other standard or non-standard extensions. Standard and Non-standard extensions may be added to any implementation as per the need of the application. In RISC-V ISA there are four standard extensions along with base integer instruction. 'M' extension is the integer multiplication and division extension. The 'A' extension is a standard atomic instruction extension. The 'A' extension adds instructions that atomically read, modify, and write memory for inter-processor synchronization. The 'F' extension is a standard single-precision floating-point extension. The 'D' extension is a double-precision floating-point extension. The base integer instruction (I) with all four standard extensions (IMAFD) is known as G (general-purpose) instruction set. Hence RISC-V 32-bit implementation RV32IMAFD may be denoted as RV32G [18,28,29].

Table 1
Comparison of instruction set architectures.

Features	MIPS	ARMv8	OpenRISC	RISC-V
Free & Open			✔	✔
Base+Ext				✔
Compressed	✔			✔
32-bit	✔	✔	✔	✔
64-bit	✔	✔		✔
128-bit				✔
Privileged ISA				✔
IEEE 754- 2008		✔		✔

A comparison of instruction set architectures are presented in Table 1. MIPS and ARMv8 are not open source. Although OpenRISC is opensource but it do not provides modular approach. As shown in Table 1 RISC-V instruction set architecture provides 32-bit, 64-bit and 128-bit implementation. It is open source and it provides modular approach (Base + extension). Also, RISC-V instruction set architecture simplifies the instruction decoding. The instruction set encoding of RISC-V 32-bit format is shown in Fig. 1. The bit location of source registers (rs1 and rs2) and destination (rd) registers in RISC-V ISA is at the same position. This simplifies the instruction decoder implementation.

The aim of this work is to develop a low-end embedded computing system. Hence, 32-bit the base integer variant of RISC-V ISA is implemented for this work. The base ISA is carefully limited to a minimum set of instructions that provide a reasonable target for compilers, assemblers, linkers, and operating systems [18,28,29]. The base integer ISA consists of integer stores, integer loads, integer computational instructions, and control-flow instructions which is sufficient to target low end applications. The architecture of proposed 32-bit single-cycle base integer implementation of RISC-V ISA (RV32I) is presented in next section.

3. Proposed RV32I micro architecture and design

This section presents the design aspects and architecture of RISC-V based single-cycle RV32I microprocessor core. The 32-bit base integer implementation of RISC-V ISA (RV32I) is a load-store architecture. In load-store architectures, only load and store instructions access memory. The ALU performs arithmetic operations only on general-purpose CPU registers. The architecture of the proposed RV32I processor is presented in Fig. 2 and discussed in this section.

The program counter is a 32-bit register that stores the address of the next instruction to be executed. When the processor starts program counter address at 0000_0000H location of instruction memory. Since both instruction and data memory shown in Fig. 2 is byte-addressable, one 32-bit instruction takes four memory locations of instruction memory. A dedicated adder is also included for incrementing the PC in each cycle. When there is no branch or jump instruction, with every positive edge of the clock, 4 is added to the program counter, and '*PC_src*' guides 'PC Control Block' to send PC + 4 to the program counter. In case of jump instruction current PC + 4 is stored in a register file. For jump

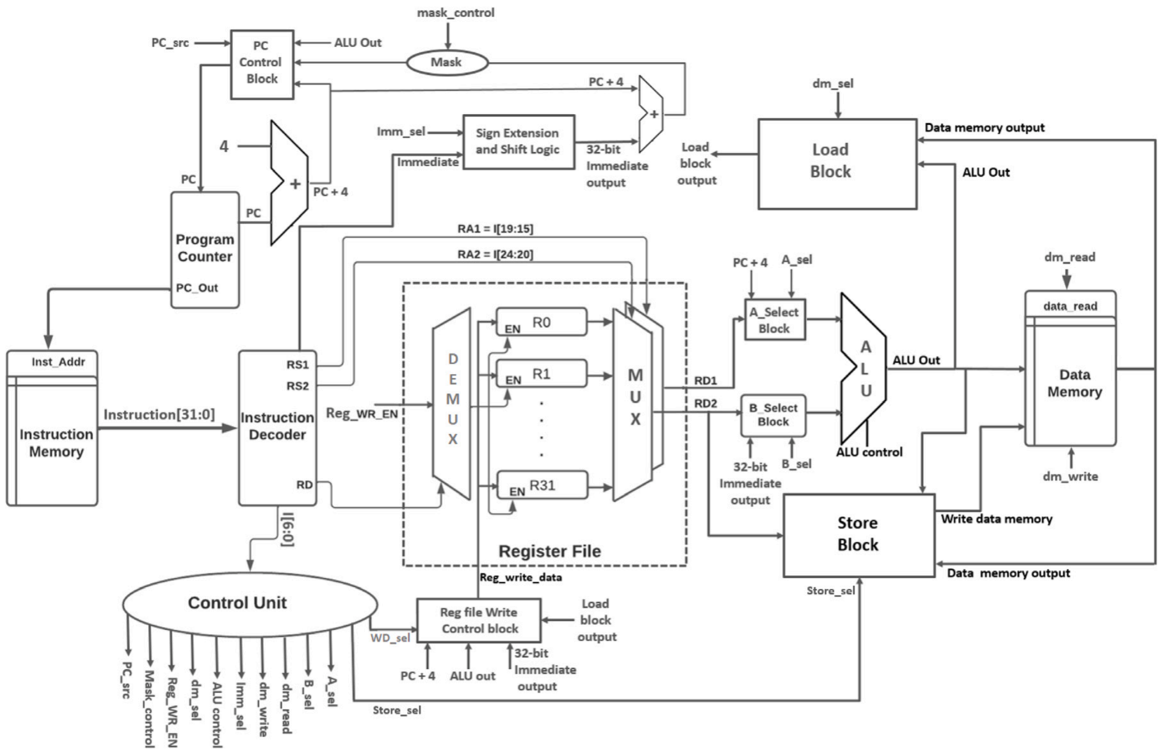


Fig. 2. Proposed RV32I processor architecture.

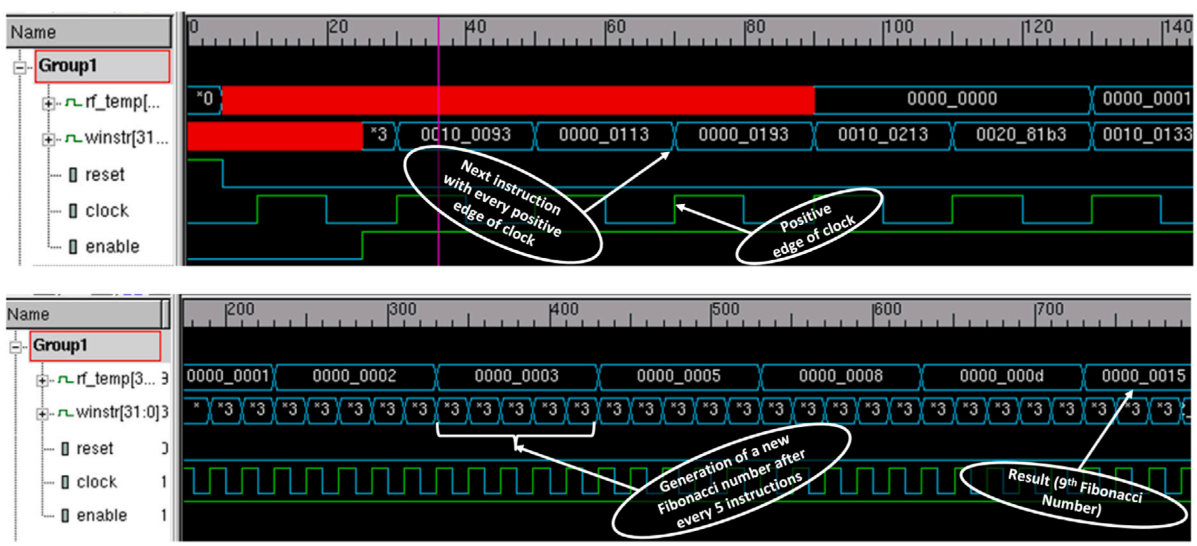


Fig. 3. Fibonacci series generation by proposed RV32I processor.

instruction, '*PC_src*' sends the 32-bit output of ALU or address calculated from the addition of 32-bit immediate and PC + 4 to program counter as shown in Fig. 2.

The Instruction decoder takes the instruction from instruction memory and sends the opcode to the control unit. Depending upon the value of opcode, the control unit generates all control signals. The control unit is purely combinational and hence, all the control signal is generated by the control unit in the same clock cycle. The register file of the RV32I processor contains 32 general-purpose registers and all are of 32-bit word size. A 5-bit address is needed to point any of the registers inside the register file. The RS1 and RS2 of the instruction decoder block shown in Fig. 2 are 5-bit long and point to source register1 and source register2 of the register file respectively. The RD of the instruction decoder block is also 5-bits long and it points to the destination register in the register file.

For R-type instruction ALU operates on two source registers and store the content in the destination register. The 'reg file write control block' sends the register write data to the register file depending upon the *'reg write data sel'* control signal generated by the control unit. The address of the register to be updated comes from the destination register address (RD) and '*reg_WR_EN*' (register write enable signal) enables that particular register to update the content.

The 'A_select Block' sends register1 data (RD1) or PC + 4 to ALU depending on the 'A_sel' signal generated from the control unit. Similarly, the B_Select Block sends register2 data (RD2) or 32-bit immediate output to ALU depending upon the value of the B_sel signal generated from the control unit. To deal with immediate value coming from

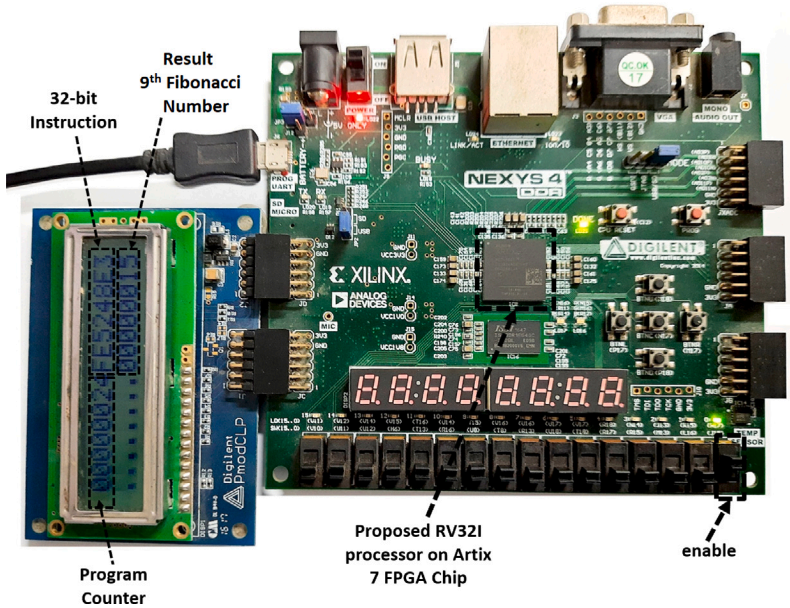


Fig. 4. FPGA prototype of RV32I processor on NEXYS DDR4 board and validation of Fibonacci series algorithm in laboratory experimental setup.

instruction the 'Sign Extension and Shift block' is used. It takes an immediate value as input from the instruction decoder and generated 32-bit immediate output.

ALU performs the arithmetic and logical operation either on two general-purpose registers of one general-purpose register and one immediate value. In load-store architecture, ALU operation is not performed on memory. The memory is accessed only through load or store instructions. ALU performs only arithmetic/logical register–register and register-immediate operations. The data memory read or write operation is performed based on 'dm_read' and 'dm_write' signals generated from the control unit. Store block stores the content of a general-purpose register to data memory depending on the 'Store_sel' signal generated by the control unit. The Load block loads the content of a memory location of data memory to a general-purpose register depending on 'dm_sel' signal that is generated by the control unit.

For the proposed processor, from the instruction fetch to write back complete operation is performed in a single clock cycle, hence the largest instruction decides the frequency of operation. To validate this processor HDL implementation, simulation, and FPGA prototype is carried and presented in the next section.

4. HDL implementation and FPGA prototype

The proposed energy-efficient single-cycle RISC-V32I processor architecture is implemented using Verilog HDL (Hardware Description Language). The functional simulation of proposed processor is presented for Fibonacci series program as an example using vcs compiler of synopsys. The proposed processor is prototyped on commercially available FPGA device with the Fibonacci series algorithm.

4.1. HDL implementation and functional simulation

The proposed architecture is implemented on Verilog HDL. An assembly program and its machine code presented in Table 2 is used as case study for functional simulation. The RTL simulation result of the proposed RV32I Processor is shown in Fig. 3. Simulation shows the generation of the Fibonacci series.

In the upper half part of Fig. 3, it is shown that the 32-bit instructions coming from instruction memory (in winstr register) are changing with every positive edge of the clock. Table 2 shows an example code snippet and corresponding machine code for Fibonacci series generation till 9th Fibonacci number by proposed RV32I processor. The value till which Fibonacci number is to be generated, is loaded in R5 ($5) of the CPU register file and Fibonacci output is taken from R3 ($3) of the CPU register file as shown in Table 2. The 9th Fibonacci number is 21, hexadecimal equivalent of 21 is 15H. In the lower half portion of Fig. 3, it can be verified that after every 5 instructions the next Fibonacci number is generated as presented in Table 2. The result of assembly program presented in Table 2 (15H) is shown in the lower half portion of Fig. 3.

Table 2
Fibonacci series generation code snippet and its corresponding machine codes.

Assembly program		Machine code
	addi $5,$0, 9	00900293H
	addi $1,$0,1	00100093H
	addi $ 2,$0,0	00000113H
	addi $3,$0,0	00000193H
	addi $4,$0,1	00100213H
Loc:	add $3, $1, $2	002081B3H
	add $2, $0, $1	00100133H
	add $1, $0, $3	003000B3H
	addi $4, $4, 1	00120213H
	blt $4, $5, Loc	FE5248E3H
HALT:	jal $31,HALT	00000FEFV

4.2. FPGA prototype and validation

For hardware level validation, the processor accommodating RV32I instructions is prototyped on a commercially available NEXYS DDR4 board containing an artix7 FPGA chip as shown in Fig. 4. For the FPGA prototype, the total delay for the single-cycle RV32I processor was reported to be 17 ns with a total on-chip power of 146 mW. The FPGA prototype of design on the NEXYS DDR4 board is able to attain a maximum frequency of 58 MHz. The FPGA utilization result of NEXYX DDR4 is shown in Table 3. The power and timing result for the FPGA prototype of the RV32I processor on NEXYS DDR4 is presented in Table 4.

Several algorithms like Fibonacci series, multiplication by addition, array load/store, and other programs were tested as part of hardware testing. However, only generation of Fibonacci series algorithm is presented here. Fig. 4 shows the hardware validation of Fibonacci series

Table 3
Resource utilization result for FPGA prototype on NEXYS DDR4 board.

Resource	Utilization	Available	Utilization %
Slice LUT	1710	63 400	2.70
Slice registers	1024	126 800	0.81
F7 muxes	269	31 700	0.85
F8 muxes	32	15 850	0.20
DSPs	0	240	0

Table 4
Delay and power result for FPGA prototype on NEXYS DDR4 board.

Delay	17 ns
Maximum frequency	58 MHz
Total on-chip power	146 mW
mW/MHz	2.51

Table 5
Synthesis report summary of RV32I targeting 180 nm CMOS process technology node.

Parameter		Value
Total area (μm^2)		243 950
Total power (mW)		14.7538
Timing report	Clock delay (ns)	20
	Maximum frequency (MHz)	50

algorithm presented in Table 2 on designed processor using NEXYS DDR4 board. LCD is interfaced to validate the result, the left half portion of first row of LCD shows the content of program counter and right half portion of first row of LCD shows the machine code of current instruction. The 2nd row of LCD shows the resulting Fibonacci series number 15H. All the number displayed on LCD is in hexadecimal format. The FPGA prototype is done to validate the design on hardware. This work aims to develop an lightweight and energy efficient microprocessor chip. Hence the proposed RV32I processor is implemented on 180 nm CMOS process technology node as part of ASIC implementation. ASIC implementation and performance details of the proposed RISC-V32I processor core architecture is discussed in next section.

5. ASIC implementation, tape-out and on-chip validation

The proposed single-cycle RV32I processor is further synthesized on the design compiler tool of Synopsys using 180 nm CMOS process technology node provided by Semi-Conductor Laboratory. At the synthesis level, the design was able to attain the maximum frequency of 50 MHz with 180 nm process technology node.

The total power consumption at the synthesis level is reported to be 14.7538 mW at 50 MHz frequency. The design takes a total cell area of 243 950 μm^2. The area, timing, and power results after synthesis of the designed processor are shown in Table 5. The gate-level synthesized netlist and Synopsys design constrain (.sdc) file from the design compiler are taken to the IC compiler tool of Synopsys for physical design. The gate level synthesized netlist and synopsys design constrain (.sdc) file from design compiler is taken to IC compiler tool of synopsys for physical design. To set the max timing slow corner is used and for the min timing fast corner is used. For timing analysis BEST_CASE for minimum delay and WORST_CASE for maximum delay have been taken. Hence, the setup is analyzed with WORST_CASE and hold with BEST_CASE.

The Input–output (IO) pads library of SCL contains input pads, output pads, tristate and bidirectional pads, clock pad and power pads. The power pad is actually a direct input pad connected to a protection device. IO pad ring of 80 pins is created keeping a pitch of 140 μm. The height of IO pad cell is 250 μm and the width is 65 μm. The height and width of corner pad cell is 250 μm. For logic input, input IO pad cells, for logic output, output IO pad cells are used in current I/O pad

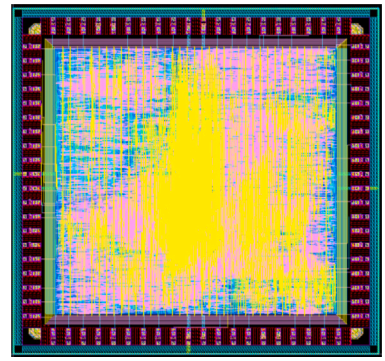
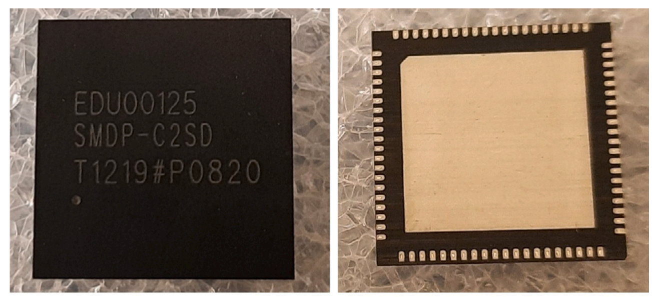


Fig. 5. Tapeout design of proposed RV32I processor.



(a) Top View (b) Bottom View

Fig. 6. Fabricated RV32I processor core chip.

frame design. The clock pad is designed to be faster and can drive high load.

The high fanout input pads, clock, reset and enable are placed near to the IO power pads. The power pad pairs for core supply is placed in the position middle of all 4 sides of ring to achieve uniform power distribution throughout the core area. The final layout of design is shown in Fig. 5. The design is able to achieve a maximum frequency of 40 MHz after physical design. After clock tree synthesis a positive skew of 1.37 ns is added. The total power consumption of the design is reported to be 11.9 mW. Since 180 nm CMOS process technology is used, the leakage power for the proposed design is very less and the dynamic power is almost equal to the total power of the design. The detailed area, power, and timing report for the physical design of the proposed RV32I architecture is presented in Table 6.

The design is able to achieve a maximum frequency of 40 MHz after physical design. After clock tree synthesis a positive skew of 1.37 ns is added. The total power consumption of the design is reported to be 11.9 mW. Since 180 nm CMOS process technology is used, the leakage power for the proposed design is very less and the dynamic power is almost equal to the total power of the design. The detailed area, power, and timing report for the physical design of the proposed RV32I architecture is presented in Table 6. It is important to note that the fabrication of this chip is done in a suttle project where the pin package and die size for the chip are fixed. Due to this reason available size of 5 mm × 5 mm die and 80L-QFN package are used for chip tapeout. The final layout of the design is shown in Fig. 5. The top and bottom view of fabricated chip is shown in Fig. 6.

The performance comparison of the designed RV32I processor core with widely used similar range commercial processor AT89C51[31] and three other processors, RATX[32], UT699[30], and GCOFP[33], is shown in Table 7. In [30] they have provided the comparison of AT89C51, RATX and UT699 processors on different parameters. In [33] it presents a domain-specific processor for image processing applications. It is designed on 0.18-μm CMOS process technology node,

Table 6
Timing report summary after physical design using 180 nm CMOS process technology node.

Timing report		Power report		Area report	
Parameter	Time	Parameter	Power	Parameter	Area
Clock delay	25.00 ns	Cell internal power	7.9336 mW	Core size	3.84 mm × 3.84 mm
Clock network delay	+1.37 ns	Net switching power	3.9058 mW	Chip size	4.74 mm × 4.74 mm
Library setup time	−0.31 ns	Total dynamic power	11.8394 mW	Die size	5.00 mm × 5.00 mm
Data arrival time	−24.80 ns	Cell leakage power	32.4361 μW	Package size	12.0 mm × 12.0 mm
Slack	1.26 ns	Total power	11.8718 mW	Pin-package	80L-QFN

Table 7
Comparison of proposed RV32I core with commercial processor cores presented in [30–33].

Parameter	AT89C51 [31]	RATX [32]	UT699 [30]	CGOFP [33]	Proposed RV32I
Clock	12 MHz	25 MHz	66 MHz	20 MHz	40 MHz
VDD	6.0 V	3.3 V	3.6 V	1.2 V	1.8 V
Process technology	–	–	–	0.18-μm CMOS	0.18-μm CMOS
Power	600 mW	500 mW	178.2 mW	36 mW	11.87 mW
MIPS	2	20	75	–	40
mW/MHz	50	20	2.7	1.8	0.29

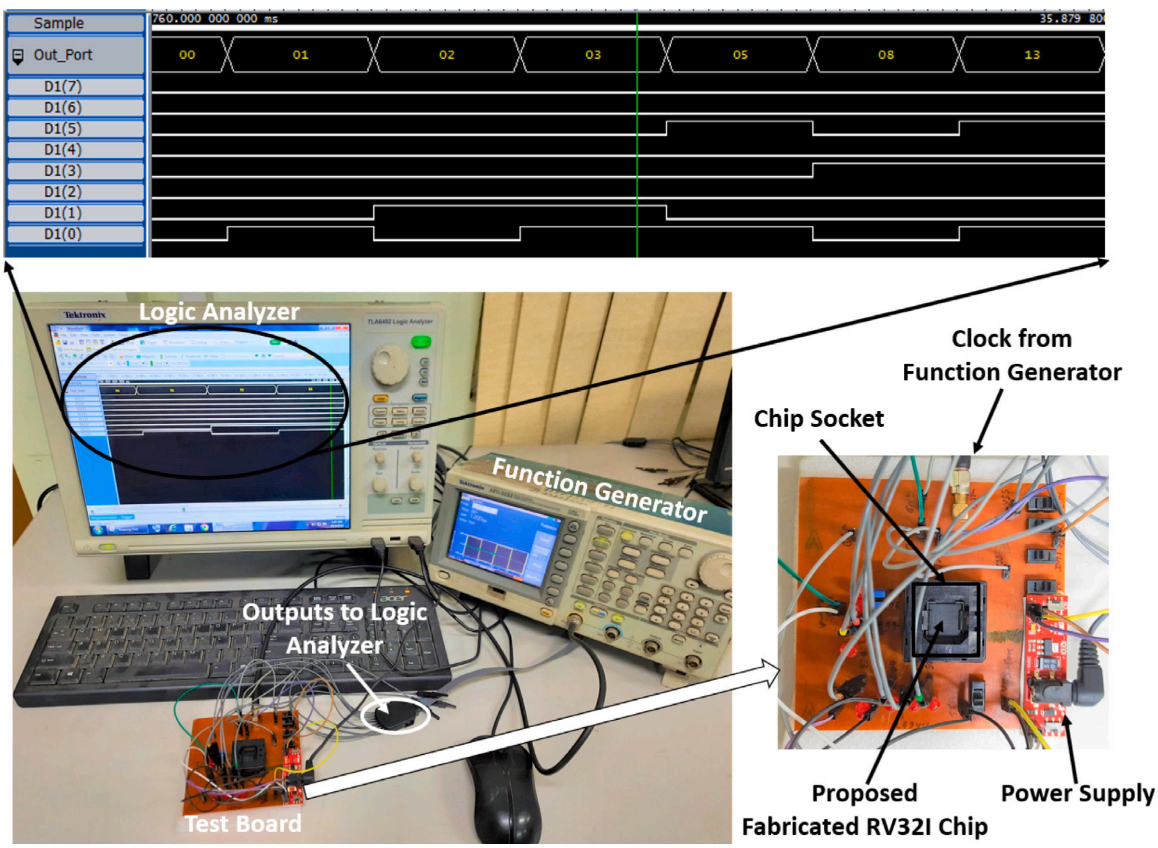


Fig. 7. Silicon-level validation of RV32I processor chip in laboratory experimental setup.

and the reported power at 20 MHz frequency is 36 mW with a 1.2 V core voltage. Although these commercially available chips are having different functional features, the proposed processor chip is having the advantage of energy efficiency. For low-end edge computing applications where the demand is to do a simpler task with low-power, the proposed processor is the best suitable candidate.

The basic test setup for silicon level validation of proposed fabricated RV32I chip is shown in Fig. 7. An in-house printed circuit board (PCB) is designed to accommodate basic test for RV32I chip. In laboratory experimental setup, for initial testing of chip, clock is supplied from function generator using SMA connector. Commercially available chip socket is used to mount the RV32I chip on PCB. The output ports are connected to TLA6402 Logic Analyzer for real time verification of output results. Fig. 7 shows the validation of Fibonacci series program on designed RV32I chip.

6. Conclusion

A Single-Cycle RISC-V ISA-based RV32I processor is designed using Verilog HDL for low-end edge computing applications. For FPGA implementation, on the NEXYS DDR4 FPGA board, the design is able to achieve 58 MHz frequency. For ASIC implementation, the design is synthesized with 180 nm CMOS process technology node. The synthesis tool reports a maximum frequency of 50 MHz for the design. After backend design, the proposed RV32I processor is able to operate on 40 MHz frequency. The chip area of the design is 5 mm × 5 mm and the total power consumption reported after the physical design is 11.87 mW. Since 180 nm process technology is used for design, the leakage power is less than 1% of total power. The proposed design is compared with commercially available conventional low-end embedded processors. The comparison result shows that the proposed RV32I processor is best suitable for low-end edge computing applications as it consumes lowest power of 0.29 mW/MHz. A basic functional testing of proposed fabricated RV32I chip is carried out in laboratory experimental setup and results are presented.

CRediT authorship contribution statement

Satyam Shukla: Conceptualization, Architecture design, RTL design, FPGA prototype, ASIC implementation, Laboratory experimental setup, Experiments and validation, Writing – original draft. **Punyesh Kumar Jha:** Design and fabrication of PCB for Chip Testing, Experiments for the chip testing. **Kailash Chandra Ray:** Conceptualization, System architecture, Design methodology, Testing plan, Supervision, Project administration, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgments

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